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Module 7 Introduction

Lesson 1: Foundations of Bayesian Regression Analysis

Lesson 2: Advanced Techniques in Hierarchical Linear Models

✔ Reading: Advanced Techniques in Hierarchical Linear Models Readings

1h

✔ Video: Hierarchical Linear Models Pt. 1

7 min

✔ Video: Hierarchical Linear Models Pt. 2

6 min

✔ Video: Hierarchical Linear Models Pt. 3

10 min

✔ Video: Hierarchical Linear Models Pt. 4

3 min

✔ Practice Assignment: Advanced Techniques in Hierarchical Linear Models Quiz

Submitted

Lesson 3: Exploring Generalized Linear Models in Bayesian Context

Module 7 Summative Assessment

Module 7 Summary

Your grade: 100%

Next item →

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Advanced Techniques in Hierarchical Linear Models Quiz

≡ Instructions

1. What is the role of residual plots in regression analysis?

1 / 1 point

✔ To assess the fit of the regression model

Jan 11, 1:04 AM IST

Attempts Unlimited

🔄 Retry

🗑️

○ To compute the posterior distribution

○ To transform the explanatory variables

○ To eliminate the need for posterior predictive checks

✔ Correct

To pass you need at least 80%. We keep your highest score.

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Residual plots are used to assess the fit of the regression model.

2. What is the purpose of using hierarchical models in Bayesian regression?

1 / 1 point

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○ To ensure the posterior distribution is proper

○ To eliminate the need for prior distributions

○ To increase the number of parameters in the model

✔ To incorporate different levels of variability and improve inference

✔ Correct

Correct! Hierarchical models incorporate different levels of variability and improve inference.

3. What is the main advantage of using a hierarchical model in Bayesian regression?

1 / 1 point

○ It ensures the residuals are normally distributed

○ It eliminates the need for posterior distributions

✔ It allows for modeling different levels of variability

○ It simplifies the model-building process

✔ Correct

Correct! Hierarchical models allow for modeling different levels of variability.

